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B.TECH
(SEM III) THEORY EXAMINATION 2022-23
CHEMICAL ENGINEERING FLUID MECHANICS

Time: 3 Hours

Total Marks: 100

Note: Attempt all sections. If require any missing data; then choose suitably.

SECTION – A

1. Attempt all questions in brief.

(2×10=20)

- (a) Define shear stress.
- (b) What is the physical significance of Reynolds number?
- (c) Define laminar & turbulent flow.
- (d) Define fanning friction factor.
- (e) Differentiate between fluid statics & fluid dynamics.
- (f) What are the assumptions of the Bernoulli's Theorem?
- (g) What is priming?
- (h) What are the disadvantages of pitot tube?
- (i) What is Dimensional analysis?
- (j) Define two phase flow.

SECTION – B

2. Attempt any three of the following:

(10×3=30)

- (a) A stream function in an X-Y plane is given by $\psi = 2xy$. Calculate the velocity at the point (2, 3). Find the velocity function Φ .
- (b) A venturimeter of throat diameter 0.1 m is fitted in the pipeline. Water at the rate of 30 lit /sec is flowing through a 0.2 m I.D. Pipe. A differential manometer in the pipeline has an indicator liquid and the manometer reading is 1.16 m. What is the relative density of the manometer liquid? Venturi coefficient = 0.96, density of water is 998 kg/m³.
- (c) The diameter of a centrifugal pump, which is discharging 0.03 m³/s of water against a total head of 20m, is 0.4m. The pump is running at 1500 rpm. Find the head, discharge & ratio of powers of a geometrically similar pump of diameter 0.25m when it is running at 3000 rpm.
- (d) Describe the working principle of manometers & give necessary expression for pressure measurement.
- (e) Give a list of various types of pipe fittings. How is the friction loss due these fittings accounted for. Explain in detail.

SECTION -C

- 3. Attempt any one part of the following: (10×1=10)**
- (a) An oil film of thickness 10mm is used for lubrication between the two square parallel plate of size 0.9 m x 0.9 m in which the upper plate moves at 2m/s requires a force of 100N to maintain this speed. Determine (i) viscosity of the oil and (ii) kinematic viscosity of oil if the specific gravity of oil is 0.95.
 - (b) Derive Hagen-Poiseuille equation and state the assumptions made.
- 4. Attempt any one part of the following: (10×1=10)**
- (a) State the Buckingham's Pi Theorem. A fluid is steadily flowing through a straight circular tube. The pressure drop P is a function of ρ , μ , v of the fluid, D and L of the tube. Develop the dimensionless groups using Pi theorem.
 - (b) A plate having an area of 0.6 m^2 is sliding down the inclined plain at 30° to the horizontal with a velocity of 0.36 m/s. there is a cushion of fluid 1.8 mm thick between the plane and plate. Find the viscosity of fluid if the weight of the plate is 280 N.
- 5. Attempt any one part of the following: (10×1=10)**
- (a) Differentiate between variable head type and variable area type flow meters. Write their advantages and limitation.
 - (b) A pitot tube is placed in the centre of a pipe of 40 cm diameter. The difference between the static and impact pressure is 6 cm water. The coefficient of discharge of pitot tube is 0.98. Determine the maximum velocity and volumetric flow rate of water.
- 6. Attempt any one part of the following: (10×1=10)**
- (a) A pipe 100 mm in diameter has a nozzle attached to it at the discharge end, the diameter of nozzle is 50 mm. the rate of discharge of water through the nozzle is 20 litre/ sec and the pressure at the base of the nozzle is 5.886 N/cm^2 . Calculate the co-efficient of discharge. Assume that the base of the nozzle and outlet of the nozzle are at the same elevation.
 - (b) Classify the various types of pumps. With the help of neat sketch, discuss the working of any reciprocating pump.
- 7. Attempt any one part of the following: (10×1=10)**
- (a) Explain characteristic curves with the help of suitable diagram. Why is priming necessary before operating a centrifugal pump?
 - (b) Define mixing and agitation. What are the various types of impellers used in chemical industries? Also discuss their selection criteria.